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Experimental and Numerical Study on Air Cooling System Dedicated to Photovoltaic Panels

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Abstract: The efficiency of solar systems, in particular photovoltaic panels, is typically low. Various environmental parameters affect solar panels, including sunlight, the ambient and module surface temperatures, the wind speed, humidity, shading, dust, the installation height, etc. Among others, the key players are indeed solar irradiance and temperature. The higher the temperature is, the higher the short-circuit current is, and the lower the open-circuit voltage is. The negative effect of lowering the open-circuit voltage is dominant, consequently lowering the power of the photovoltaic panels. Passive or active cooling systems can be provided to avoid the negative effect of temperature. This paper presents a prototype of an active cooling system dedicated to photovoltaics. The prototype of such a system was developed at the AGH University of Kraków and tested under laboratory conditions. The proposed system is equipped with air fans mounted on a plate connected to the rear part of a 70 Wp photovoltaic panel. Different configurations of the system were tested, including different numbers of fans and different locations of the fans. The artificial light source generated a irradiation value of 770 W/m^2. This value was present for every variant tested in the experiment. As observed, the maximum power generated in the photovoltaic panel under laboratory conditions was approx. 47.31 W. Due to the temperature increase, this power was reduced to 40.09 W (when the temperature of the uncooled panel surface reached 60 °C). On the other hand, the power generated in the photovoltaic panel equipped with the developed cooling system was approx. 44.37 W in the same conditions (i.e., it was higher by 10.7% compared to that of the uncooled one). A mathematical model was developed based on the results obtained, and simulations were carried out using the ANSYS Workbench software. After the validation procedure, several configurations of the air cooling system were developed and analyzed. The most prominent case was chosen for additional parametrical analysis. The optimum fan orientation was recognized: a vertical tilt of 7° and a horizontal tilt of 10°. For the tested module, this modification resulted in a cost-effective system (a net power increase of ~3.1%).

Keywords: solar energy; photovoltaics; PV; air cooling of PV panels; parametric optimization; CFD

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1. Introduction

The modern world is becoming increasingly aware of the need to use renewable energy sources (RESs) to meet the growing demand for electricity. The use of RESs contributes to reducing the negative impact on the environment, and, above all, plays an important role in decarbonization [1]. The main advantage of an RES is that not only is heat obtained (biomass and solar collectors), but it also generates electricity directly (photovoltaic panels and wind turbines), without the need for energy conversion, as is the case when, e.g., burning fossil fuels. Solar energy, especially PV (photovoltaic) installations, plays an increasingly important role in the energy transition because it is relatively cheap, and, above all, it is clean and increasingly efficient [2]. Compared to the previous years, there has been a significant increase in the installed capacity and energy produced by PV panels. In the case of Poland, the increase in the PV installed capacity was more than 30% in 2023,

with more than 17 GW of PV panels currently installed [3]. However, the performance of photovoltaic panels is closely related to their operating temperature [4]. The higher the temperature is, the higher the short-circuit current is, and the lower the open-circuit voltage is. Consequently, an increase in the PV panel surface temperature can decrease the solar-to-electricity conversion efficiency, which is a significant challenge, especially in regions with high ambient temperatures or during heat waves. Therefore, developing effective cooling systems for PV panels is crucial to ensure their optimal performance and long-term operation. In addition, cooling PV panels can increase the longevity of the panels by reducing the risk of material degradation caused by high temperatures [5].

Various methods of cooling PV panels have been researched and described in the literature worldwide, each with their advantages and disadvantages, but they may not always be applicable in all areas. There are various subdivisions of cooling methods, but it is more accurate to define them as active and passive methods for cooling PV panels [6]. Active systems use a cooling medium (e.g., water or air), and for the cooling process to take place, the cooling medium must be moved by a pump or fan, which requires additional energy consumption [7]. Passive cooling systems do not use additional equipment, so such systems may be less complex, but their efficiency is usually lower than that of active systems [8]. Water cooling uses a water flow to dissipate excess heat. Sprinkler systems can be used on the front or rear of the PV panels. In this case, water consumption must be considered, which may not be negligible. Odeh and Behnia [9] conducted an experimental study of the water cooling of the front surface of PV panels together with a water collection tank. The work carried out resulted in a 15% increase in the efficiency of the cooled panel compared to that of the uncooled panel. Sornek et al. [10] conducted experimental work comparing the performances of PV panels with different water cooling capacities of 50 Wp and 310 Wp. The proposed solution resulted in a 10% increase in the power output compared to that of the uncooled panels. In contrast, another paper [11] investigated the effect of water flow and the placement and diameter of water spray nozzles. The experimental work achieved an increase in efficiency of 13% and an increase in energy production of more than 10% compared to those of the uncooled PV system. Another design of active cooling systems is hybrid systems combining a PV panel and a solar collector in a single unit (PV/T systems), which allows for much higher efficiencies to be achieved by generating electricity and heat simultaneously [12]. Nahar et al. [13] investigated a water-based PV/T system with a novel pancake-shaped flow channel. The proposed solution achieved a thermal efficiency of 50–60% and an electrical efficiency of 10%. Systems using thermoelectric modules are also used. These are placed in direct contact with PV panels to collect heat from their surface and convert it into electricity [14]. There is also a more unconventional approach to increasing the efficiency of photovoltaic systems in combination with thermal systems, spectral splitting. The radiation spectrum is separated into the infrared part used in thermal systems and visible and near-infrared radiation for photovoltaics [15]. Zhang et al. [16] investigated the use of thermoelectric generators (TEGs) in PV/T collectors. Numerical analysis and experimental studies were carried out. When thermoelectric generators were used, it was possible to obtain almost 3% more energy than that of the system without TEGs. It is also possible to use phase change materials (PCMs), which can absorb excess heat from the surface of PV panels so their temperature can be stabilized [17]. In this case, the surface temperature of a PV panel can be controlled by selecting a suitable material with the required phase change temperature. The systems that may be increasingly used, due to their potential high efficiencies, are concentrated photovoltaic (CPV) systems. Unfortunately, their use is associated with the excessive heating of the photovoltaic cells. In this case, it is important to develop small, but very efficient heat sinks in order not to exceed the temperature limits for the solutions used [18]. Another interesting approach to the application of mainly high-power photovoltaic systems is floating photovoltaic systems. By placing the PV systems on water, it is possible to reduce the surface temperature of the PV panels (increasing efficiency), and at the same time,

impede the evaporation of water from water basins (the possibility of having more water available for irrigation or use in hydroelectric power plants) [19].

The most commonly used cooling method is air cooling using natural convection. This is the simplest method and involves an airflow in the space between the PV panel and the surface on which the PV panels are mounted (e.g., ground or roof) [20]. Passive cooling can also be achieved by using a special design of heat sinks [21], fins [22], or the support structure of the PV panels [5], which increase the efficiency of heat removal from the panel surface, and consequently reduce the PV panel temperature. Another air cooling technology is active cooling, i.e., using fans or blowers to force airflow within the PV panel, causing it to lower its temperature. Patil et al. [22] presented an experimental study of a backside air-cooled PV panel and a reference panel without cooling. The efficiency and power output of the PV panels were investigated for different airflow rates. The study found that air cooling increased the efficiency of the PV panel from 7% to 12.6%. Elminshawy et al. [23] experimentally investigated the possibility of using pre-cooled air in the ground to cool PV panels. Measurements were made for different airflow rates and air temperatures. Using such a system increased the average efficiency of the cooled PV panel by almost 30% compared to that of the uncooled panel. In addition to experimental work, computational fluid dynamic (CFD) simulations are used to develop numerical models to optimize cooling systems. Hussien et al. [24] investigated the possibility of using small fans to cool the backside of PV panels, increasing the power generation efficiency by 2.1%. The CFD model allowed for the determination of PV panel surface temperatures, and good agreement was reached between the model and the experiment.

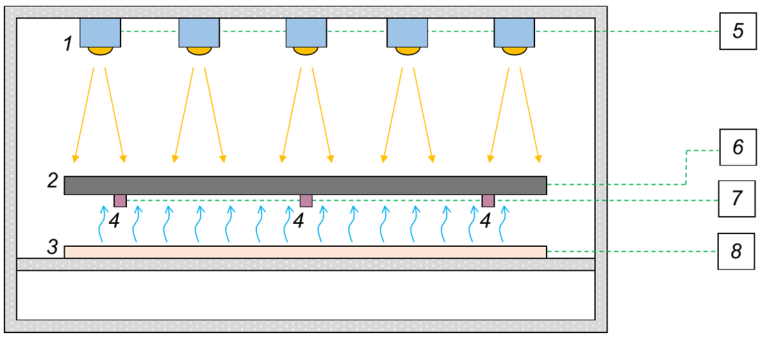
This work is a continuation of the studies described in Refs. [10,11], where the initial water and air cooling system configurations were considered. The work carried out in the following research shows a significantly different approach to PV panel cooling than previously proposed. Due to the need to conserve water and the increasing problem of access to water, the possibility of using other media to cool the surface of PV panels should be considered. Such a medium could be air, which, with the right channeling and flow velocity, could be worth considering. The contribution of this work is mainly related to developing prototypical air-cooling plates, which can be integrated with PV panels' mounting systems and used to decrease the panels' operating temperature. The proposed system is equipped with low-power air fans to provide efficient cooling with the low self-consumption of electricity. Such a solution can be useful, especially when PV panels operate in hot climates and during summer in other climates.

2. Materials and Methods

2.1. Experimental Rig

An experimental rig was used to investigate the impact of temperature on the operational parameters of the tested PV panel and the possibility of improving the panel's performance using a dedicated cooling system. The rig has horizontal dimensions of 1.25×0.75 m and is equipped with 20 halogen lamps (see Figure 1). This type of light source is widely used as an infrared light source in multi-source solar simulators [25]. Comparing the spectral distribution of halogen and sun radiation (available, for example, in Ref. [26]), it can be noted that halogen is best suited in the range of 550 to 750 nm. Furthermore, natural sunlight has a color temperature of approx. 5600 K, whereas halogen lamps radiate at a black-body temperature of about 3200 K. As a result, they radiate less in the shorter wavelengths, but more in the infrared portion [27]. In the discussed experiments, the error in the spectrum compared to sunlight is ignored because the authors did not measure the output as a function of wavelength, but compared it as a function of fan number and their location for PV panel cooling.

(**a**)



(**b**)

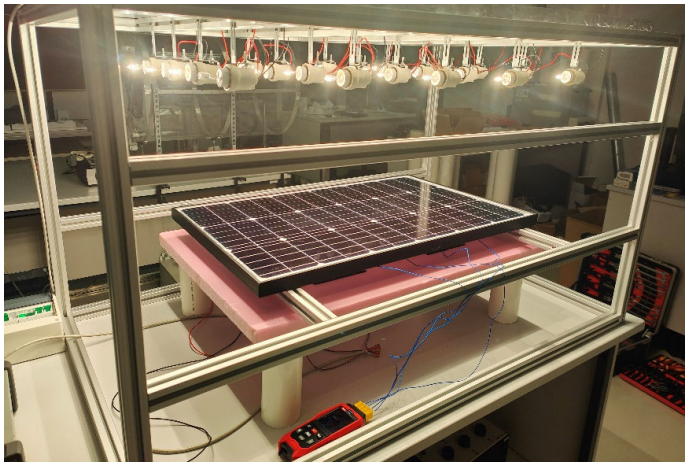


Figure 1. Experimental rig—scheme (**a**) and real view (**b**). 1—halogen lamps; 2—PV panel; 3—air cooling plate; 4—temperature sensors; 5—light switch; 6—data logger; 7—electronic load; 8—air fans switch.

A dedicated cooling system was designed in the form of cooling plates fitted with air fans (two, three, or six, depending on the series). The air fans have dimensions of 14 × 14 cm and generate airflow at a level of 1.7 m/s. Each fan has a rated power of approximately 2.4 W, so the cooling system power requirement ranged from 4.8 to 14.4 W, depending on the tested configuration. The distance between the PV panel and the plate with the integrated air fans is 10 cm. This value relates to the distance between the PV panel and the roof in actual PV installations. Different values can be observed for different mounting systems, but 10 cm can be considered as the average. Figure 2 shows the configurations of the cooling plates used in the experiments discussed.

(**a**)

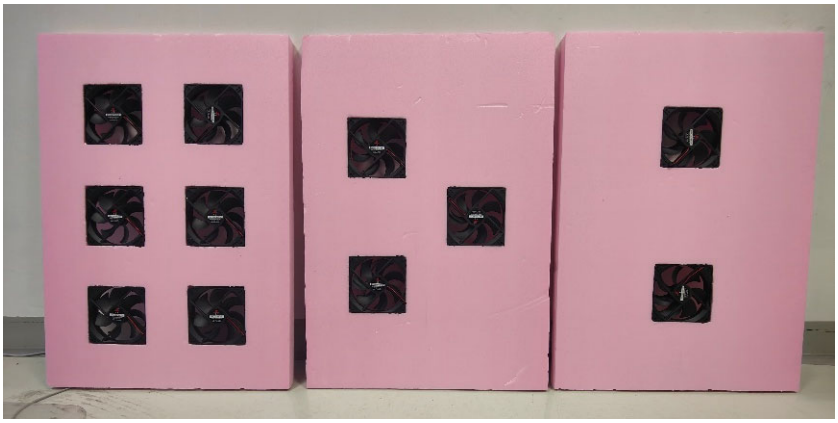


Figure 2. *Cont.*

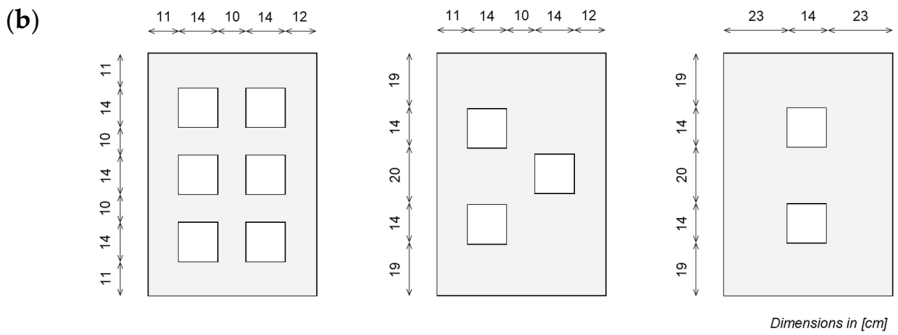


Figure 2. Configurations of the cooling plates used in the discussed experiments (**a**) and their dimensions (**b**).

Two-state relays controlled the operation of the light source and the cooling system. Three thermocouple sensors located on the PV panel's rear side measured its surface temperature. Another thermocouple sensor measured the temperature of the ambient air. The PV panel was connected to the electronic load, which operated in the constant current (CC) mode and was used to simulate different operating conditions. Finally, a hand-held pyranometer was used to determine the average light intensity on the front surface of the PV panel. The main parameters of the measurement components used in the tests carried out are shown in Table 1.

Table 1. The main parameters of the components used during the tests.

Element	Description
PV panel	Monocrystalline panel (4SUN 70W Maxx) characterized by the following: • Maximum power (P_{max}) = 70 Wp; • Current at P_{max} (I_{mp}) = 3.88 A; • Voltage at P_{max} (V_{mp}) = 18.00 V; • Short-circuit current (I_{sc}) = 4.19 A; • Open-circuit voltage (U_{oc}) = 21.60 V; • Temperature coefficient of power ($P_{\%}$) = −0.38%/°C; • Dimensions of 0.81 × 0.51 × 0.03 m.
Light source	20 halogen lamps with electrical power of 150 W each
Electrical load	Electronical load ARRAY 3721A with the following input ratings: • Current 0–40 A (accuracy 0.05% + 8 mA); • Voltage 0–80 V (accuracy 0.1% + 8 mV); • Power 0–400 W (accuracy 0.1% + 600 mW).
Pyranometer	The unit PYR20 with a measuring range from 0 to 2000 W/m^2, spectral range from 400 to 1100 nm, and accuracy of 5%
Temperature sensors	K-type thermocouple sensors (NiCr-NiAl) with a measuring range from −50 °C to 400 °C and tolerance ±2.0 °C or ±0.0075 × [t], where t is the measured temperature

2.2. Experimental Procedure

During the discussed studies, the PV panel without a cooling system was tested in series S_1, and the PV panels equipped with three configurations of the air cooling system were tested in series S_2–S_4:

- series S_2: air cooling system with cooling plate equipped with six air fans;
- series S_3: air cooling system with cooling plate equipped with three air fans;
- series S_4: air cooling system with cooling plate equipped with two air fans.

The above configurations were tested in different operating conditions (cases *A*, *B*, and *C*). In each case, the different values of the uncooled PV panel temperature were assumed as reference conditions:

- case *A*: PV panel temperature similar to air temperature in the laboratory;
- case *B*: average PV panel temperature equal to approx. 50 °C;
- case *C*: average PV panel temperature equal to approx. 60 °C.

2.3. CFD Model Description

2.3.1. Geometry and Mesh of Computational Domain

A CFD model was created to take a closer look at the heat transfer processes in the cooled PV panel and to perform variational and parametric analyses of the proposed system. The ANSYS Workbench (Wb) environment was used for this work. The geometry of the model was created in Wb's native program, SpaceClaim. It mainly included the volume of air underneath the real-size PV panel. The same geometry was used for all the options considered in the following study. The created body is shown in Figure 3, with the domain boundaries highlighted. The prepared numerical model had to be adapted for subsequent variational and parametric analyses. For this reason, the geometry was planned in such a way that the number and layout of fans could be modified quickly. Six potential fan locations were assumed (see Figure 4).

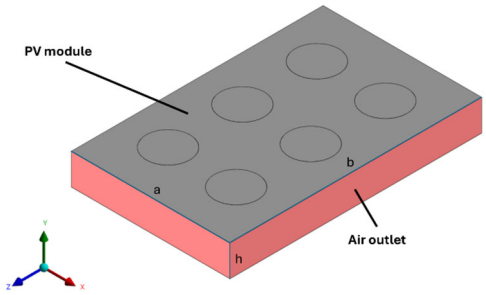


Figure 3. Geometry of computational domain (dimensions: a = 0.51 m, b = 0.81 m, and h = 0.1 m).

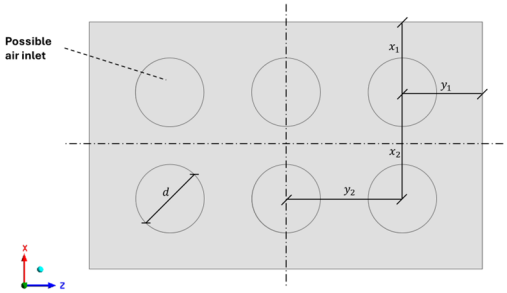


Figure 4. Bottom side of computational domain (dimensions: x_1 = 0.145 m, x_2 = 0.23 m, y_1 = 0.165 m, y_2 = 0.24 m, and d = 0.12 m).

The PV panel is represented as an infinitely flat wall. This significantly reduces the number of cells in the computational mesh and increases the speed of calculations, which is important for multivariate analysis. The mesh was created using the native ANSYS Workbench 2023 R2 software, ANSYS Meshing (see Figure 5). It consisted of 778,890 cells and had high-quality coefficient values: an average orthogonal quality of 0.993 and an average skewness of 0.072. Regarding the aspect ratio, the values were quite high near the surface of the PV model, with the highest value being 26. This is due to the addition of an inflation layer near this surface. Its presence is required because of the implementation of the k-ω turbulence model. Lowering the aspect ratio was possible. Unfortunately, it would have required an increase in the number of grid cells, which would have made the calculation considerably longer. The model was designed for parametric analyses, where it is worth making some simplifications in exchange for faster results. Depending on the source of the data, the aspect ratio can be a value ranging from 20 to even 40 (higher

than 1000 in specific situations) [28]. Therefore, this is not a significant deviation from the recommendations.

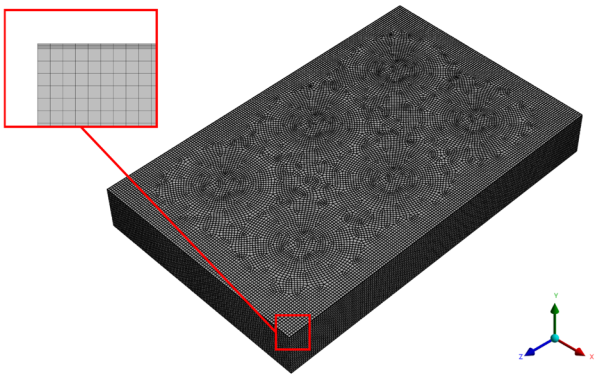


Figure 5. The mesh used in the CFD model.

2.3.2. Mathematical Model and Solver Setup

The calculations were carried out using the ANSYS Fluent software (2023 R2). It is a solver based on a series of governing equations [28]: continuity (Equation (1)), the conservation of momentum (Equations (2)–(4)), and the conservation of energy (Equation (5)). For the last one, an additional source term S_r [W/m^3] was included, which represented the value of the heat supplied to the PV panel associated with the influence of solar radiation.

$$\frac{\partial \rho}{\partial t} = \left(\frac{\partial \rho u}{\partial x} + \frac{\partial \rho v}{\partial y} + \frac{\partial \rho w}{\partial z} \right) = 0 \quad (1)$$

$$\rho \frac{Du}{Dt} = \frac{-\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) \quad (2)$$

$$\rho \frac{Dv}{Dt} = \frac{-\partial p}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) \quad (3)$$

$$\rho \frac{Dw}{Dt} = \frac{-\partial p}{\partial z} + \mu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) \quad (4)$$

$$\rho \frac{DT}{Dt} = \frac{\lambda}{C_p} \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) + S_r \quad (5)$$

where

ρ—the density of fluid $\left[\frac{\text{kg}}{\text{m}^3}\right]$;
t—time [s];
u, v, w—components of the velocity vector $\left[\frac{\text{m}}{\text{s}}\right]$;
T—fluid temperature [K];
λ—the thermal conductivity of the fluid $\left[\frac{\text{W}}{\text{m}\cdot\text{K}}\right]$;
C_p—the specific heat of the fluid $\left[\frac{\text{J}}{\text{kg}\cdot\text{K}}\right]$.

The PV panel was implemented as a thin wall with shell conduction assumptions. This means that the thickness of the layers was not included in the geometry, but only in the numerical calculations in ANSYS Fluent itself. The material parameters of the PV panel were entered analogously to the assumptions in Ref. [24] and may have a small discrepancy with the panel used in the experiment. Notably, the antireflective coating and the rear metal contact layer were disregarded due to their small thickness. The modeled PV panel was assumed to consist of a glass layer, protective Ethylene Vinyl Acetate (EVA) film on both sides of the PV cell, and electro-insulating Polyvinyl fluoride (PVF) film. These should be considered standard materials that allow for adequate convergence despite their potential

inconsistency with the actual design of the panel being experimentally tested. According to Ref. [29], it is possible to relate the heat generation flux for each layer to the solar irradiance value G [W/m^2]. This procedure allowed for the assignment of constant source term values separately for each component of the PV panel. Equation (6) was used for this purpose. It shows the dependence of heat generation flux q_i [W/m^3] on the efficiency of the PV panel η_{SC} [-], as well as the following physical parameters of each layer: the surface area exposed to solar radiation A_i [m^2] and the material volume V_i [m^3]. The following optical properties are also considered: the absorptivity α_i [-] and the transmissivity of the layers above τ_i [-]. The index i stands for the individual layer of the PV panel. These consider the experimentally determined irradiance of G = 770 W/m^2 and the electrical efficiency of the PV panel of η_{SC} = 0.165.

$$q_i = \frac{(1-\eta_{sc})\cdot G\cdot\alpha_i\cdot\tau_i\cdot A_i}{V_i} \quad (6)$$

Heat transfer by natural convection and radiation was assumed on the panel's surface. This was achieved using a convective heat flux application (Equation (7)) on the solid wall corresponding to the PV panel. The magnitude of this flux is calculated considering the heat penetration coefficient α_C [W/(m^2·K)] determined using Nusselt correlation. The radiative heat flux Q_R [W] was calculated using the Stefan–Boltzmann law (Equation (8)). The implementation of this phenomenon and the other boundary conditions are provided in Table 2.

$$Q_C = \alpha_c\cdot(T_A - T_\infty) \quad (7)$$

where T_A and T_∞ are, respectively, the surface temperature of the body washed with fluid and the ambient temperature [K].

$$Q_R = \varepsilon\cdot\sigma\cdot\left(T_R^4 - T_\infty^4\right) \quad (8)$$

where:

$\sigma = 5.67\cdot10^{-8}\frac{\mathrm{W}}{\mathrm{m^2\cdot K^4}}$—Stefan–Boltzmann constant;
ε—emissivity of the material [-];
T_R—surface temperature [K].

Table 2. Selected parameters of the boundary conditions used in the model.

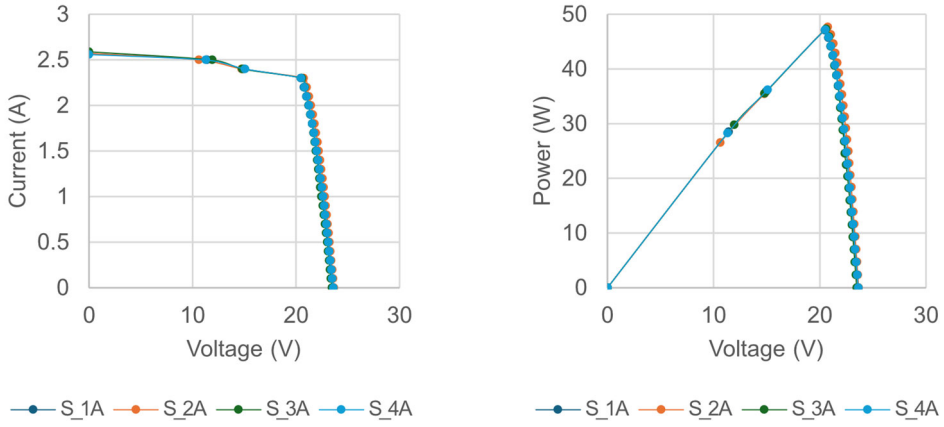
Location	Boundary Condition	Definition
Air inlets	Velocity inlet	Fixed value air velocity v = 1.7 m/s
Air outlets	Pressure outlet	Fixed value of ambient pressure p = 101,325 Pa; Reversed flow temperature T = average temperature on outlet.
Wall near inlets	Adiabatic wall	Adiabatic conditions heat flux Q = 0 W
PV panel	Wall, mixed heat transfer	$\alpha_c = f(Pr, Gr) = 6.25\left[\frac{\mathrm{W}}{\mathrm{m^2\cdot K}}\right]$; Emissivity $\varepsilon = 0.92$; $T_\infty = 293.15$ K; Shell conduction of 5 layers with heat generation rate in each one.

3. Results and Discussion

3.1. Experimental Results

The experimental part of the presented investigations was carried out to evaluate the practical possibility of cooling PV panels with low-power fans. Furthermore, the selected results obtained in this phase were used as the basis for the validation of the CFD model. As observed, the average irradiance was equal to 770 W/m^2. The maximum obtained

power varied from 47.08 ± 0.65 W to 47.68 ± 0.65 W when the average temperature of the PV panel surface was around 22.4–24.7 °C (depending on the series). It should be noted that the power generated under laboratory conditions corresponds to approx. 67.6% of the nominal power of the tested panel under STC. However, the maximum difference between the subsequent series does not exceed 0.60 W (equal to 1.3% of the maximum observed power). Thus, the testing conditions can be considered stable during all the discussed series. The operating characteristics of the tested PV panel during series S_1A–S_4A and the main important results are presented in Figure 6 and Table 3.



(**a**) (**b**)

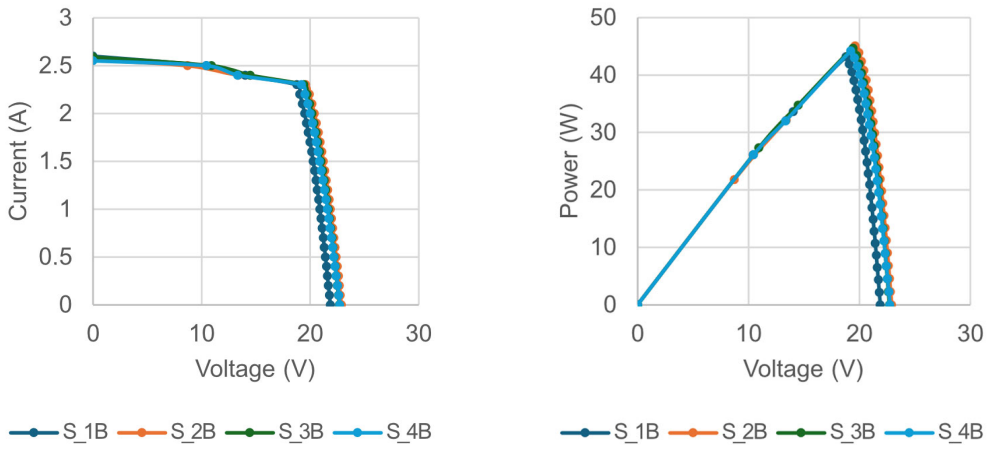
Figure 6. The characteristics of the tested PV panel during series S_1A–S_4A: (**a**) current–voltage; (**b**) power–voltage.

Table 3. The crucial data obtained in series S_1A–S_4A.

Parameter	Series S_1A	Series S_2A	Series S_3A	Series S_4A
Open-circuit voltage V_{OC} [V]	23.60	23.66	23.47	23.62
Short-circuit current I_{SC} [A]	2.57	2.57	2.59	2.56
Matched power P_{MPP} [W]	47.08	47.68	47.31	47.17
The average value of matched power P_{MPP} [W]	47.31			
Voltage at P_{MPP} V_{MPP} [V]	20.47	20.73	20.57	20.51
Current at P_{MPP} I_{MPP} [A]	2.30	2.30	2.30	2.30

On the other hand, when the PV panel was operated without a cooling system, its surface temperature was significantly higher; during the described tests, the maximum PV panel temperature exceeded 70 °C in the laboratory conditions (however, this value was higher than the temperature that can be obtained in real conditions, so it was not considered in further investigations). Series S_1B was conducted when the average temperature of the uncooled PV panel reached approximately 50 °C (the average irradiance was equal to 770 W/m^2). The power generated in such a situation was 43.19 ± 0.64 W; this value was lower by 9.1% compared to the average previously observed (47.31 ± 0.65 W). The next step was devoted to determining the impact of introducing the cooling system on the performance of the tested PV panel (series S_2B–S_4B). The introduction of the air cooling system reduced the temperature to 36.4–38.3 °C (depending on the series). When the cooling system was installed, the power ranged from 44.18 ± 0.64 W (series S_4B) to 45.06 ± 0.65 W (series S_2B). Compared to the value obtained in series S_1B, the increase in

generated power was 2.2%, 3.4%, and 4.3%, respectively, in series S_4B, S_3B, and S_2B. The I-U and P-U characteristics of the tested PV panel during series S_1B–S_4B are shown in Figure 7, and the most important results are summarized in Table 4.



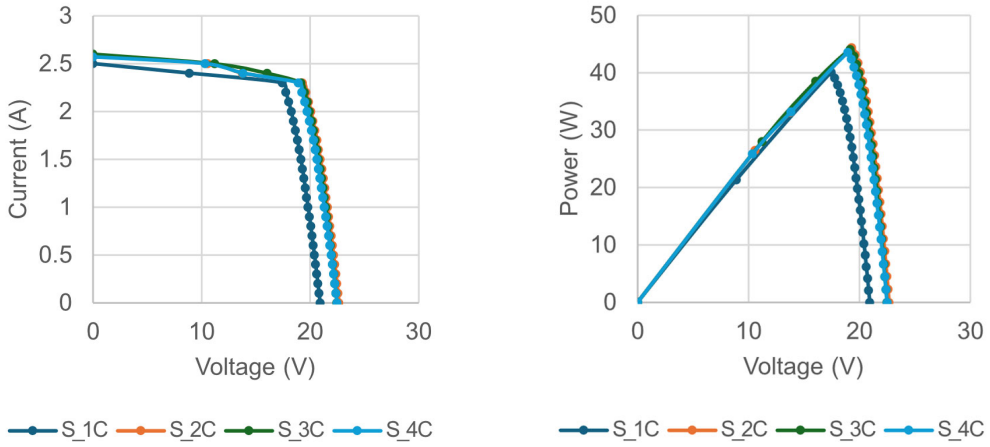
(**a**) (**b**)

Figure 7. The characteristics of the tested PV panel during series S_1B–S_4B: (**a**) current–voltage; (**b**) power–voltage (series S_1B—PV panel without a cooling system and series S2B–S_4B—PV panel equipped with three configurations of the air cooling system).

Table 4. The crucial data obtained in series S_1B–S_4B.

Parameter	Series S_1B	Series S_2B	Series S_3B	Series S_4B
Open-circuit voltage V_{OC} [V]	21.87	22.89	22.70	22.74
Short-circuit current I_{SC} [A]	2.60	2.56	2.59	2.55
Matched power P_{MPP} [W]	43.19	45.06	44.67	44.18
Voltage at P_{MPP} V_{MPP} [V]	18.78	19.59	19.42	19.21
Current at P_{MPP} I_{MPP} [A]	2.30	2.30	2.30	2.30
Decrease in the PV panel temperature ΔT_{PV} [K]	---	16.5	14.6	14.0
Increase in power generation ΔP [%]	---	4.3	3.4	2.2

Finally, the last measurements were taken when the average temperature of the uncooled PV panel reached approximately 60 °C (the average irradiance was equal to 770 W/m^2). In this case, the maximum power generated in the uncooled PV panel was 40.09 ± 0.64 W (it was lower by approx. 15.9% compared to the maximum power observed under laboratory conditions). The power generated was significantly higher in the series S_2C–S_4C (when the air cooling system was active). In the series S_2C, it was 44.37 ± 0.64 W (approx. 6.9% lower than the maximum value), while in the series S_3C and S_4C, it was 44.07 ± 0.64 and 43.54 ± 0.64 W, respectively. The operating characteristics of the tested PV panel during the series S_1C–S_4C are shown in Figure 8, and the most important results are included in Table 5.



(**a**) (**b**)

Figure 8. The characteristics of the tested PV panel during series S_1C–S_4C: (**a**) current–voltage; (**b**) power–voltage (series S_1C—PV panel without a cooling system and series S2C–S_4C—PV panel equipped with three configurations of the air cooling system).

Table 5. The crucial data obtained in series S_1C–S_4C.

Parameter	Series S_1C	Series S_2C	Series S_3C	Series S_4C
Open-circuit voltage V_{OC} [V]	20.92	22.66	22.47	22.48
Short-circuit current I_{SC} [A]	2.50	2.58	2.50	2.57
Matched power P_{MPP} [W]	40.09	44.37	44.07	43.54
Voltage at P_{MPP} V_{MPP} [V]	17.43	19.29	19.16	18.93
Current at P_{MPP} I_{MPP} [A]	2.3	2.3	2.3	2.3
Decrease in the PV panel temperature ΔT_{PV} [K]	---	24.1	20.5	18.7
Increase in power generation ΔP [%]	---	10.7	9.9	8.6

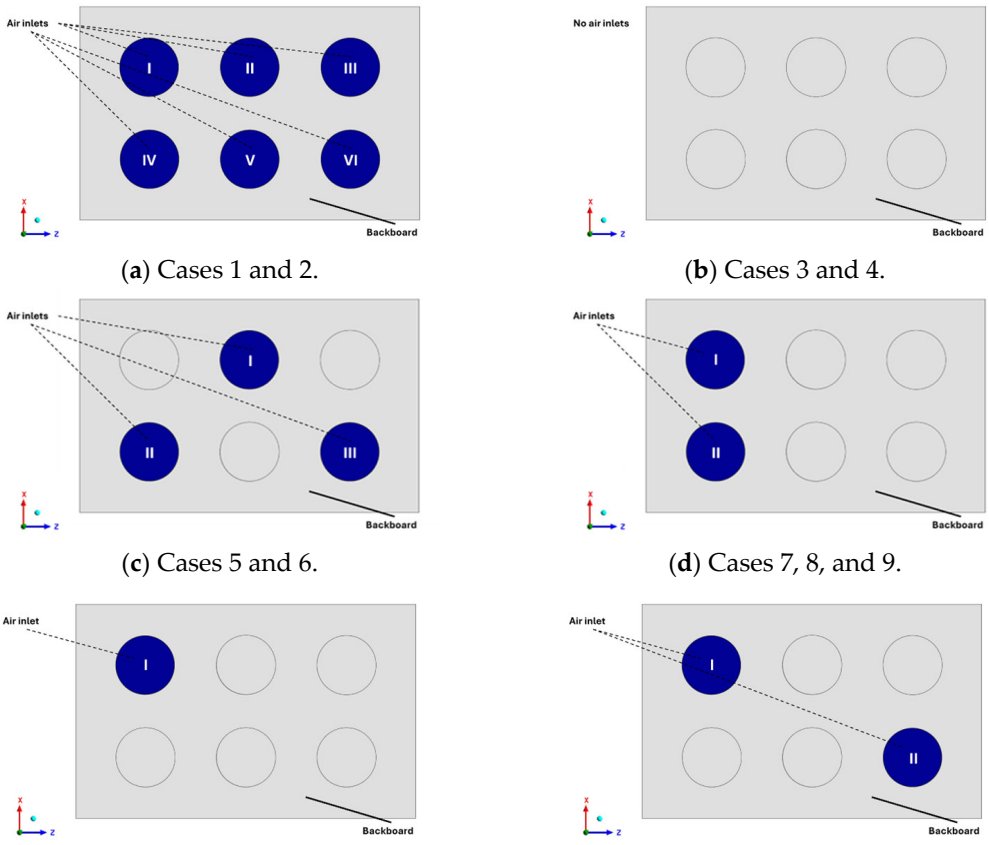
Comparing the obtained results, it can be observed that there is a significant difference between the operation parameters of the uncooled and cooled PV panels. The maximum power observed during series S_2B (when the PV panel operated with the cooling plate equipped with six air fans) was higher by 4.3% compared to that of series 2_2A (uncooled PV panel). Respectively, considering series S_3B and S_3B, this difference was approx. 10.7%. It can be concluded that during summer conditions, especially in hot climates, the introduction of cooling plates increases the power generation in the PV panel by at least 10%. On the other hand, comparing the series with an installed cooling system, the difference between the considered variants is small. Consequently, it is possible to achieve the efficient cooling of the tested PV panel using only two air fans. This information can be useful for developing future versions of the proposed system.

3.2. CFD Case Description

The results of the experimental work on the possibility of cooling the surface of the PV panels presented in Section 3.1 showed the possibility of increasing energy production. The use of this type of technology may be economically viable, but in order to obtain an

optimal solution, it would be necessary to carry out a series of experimental studies taking into account the different placements of fans, their number, the angle of inclination, and the distance from the PV panel surface. It should be borne in mind that experimental work would take a very long time, and it may not be possible to find an optimal solution. At present, it is possible to rely not only on experiments, but also on CFD modelling, which can select a suitable, optimized configuration. Based on the experimental studies, a universal numerical model was created and validated. On this basis, a series of simulations were carried out, as described below.

First of all, variational analysis was chosen. A total of 11 cases were created and analyzed (see Figure 9). It is worth noting that during the tests, not only the distribution and number of fans changed, but also the inclination of the PV panel itself and the angle of the attack of the airflow on its surface. The first analyzed cases (see Figure 9a–d) were analyzed for the effect of tilting the panel to the ground at an angle of 30°. This value in Poland is considered beneficial for an actual photovoltaic installation. This procedure served as an additional verification of the validity of the experimental studies for the horizontal orientation of the PV panel. It is worth noting that Cases 3 and 4 (see Figure 9b) assume that the PV panel is not affected by the fans, and thus the main heat transfer process is natural convection. For the last two fan settings (see Figure 9d,e), an additional assumption was made about the inclination of the direction of the airflow with respect to the panel surface. An angle of 60° was used, guided by the possibility of maximizing intensified heat transfer. All the cases that were tested are summarized in Table 6.



(**a**) Cases 1 and 2.

(**b**) Cases 3 and 4.

(**c**) Cases 5 and 6.

(**d**) Cases 7, 8, and 9.

(**e**) Case 10.

(**f**) Case 11.

Figure 9. The presentations of different variants of the fan layout under the PV panel, along with an indication for which case they were used.

Table 6. Description of all analyzed cases.

Case Number	Number of Air Inlets	Panel Tilt	Fan Tilt
1 (Figure 9a)	6	No	No
2 (Figure 9a)	6	Yes (30°)	No
3 (Figure 9b)	0	No	-
4 (Figure 9b)	0	Yes (30°)	-
5 (Figure 9c)	3	No	No
6 (Figure 9c)	3	Yes (30°)	No
7 (Figure 9d)	2	No	No
8 (Figure 9d)	2	Yes (30°)	No
9 (Figure 9d)	2	Yes (30°)	Yes (60° in relation to the axis of the panel)
10 (Figure 9e)	1	Yes (30°)	Yes (60° in relation to the axis of the panel)
11 (Figure 9f)	2	Yes (30°)	Yes (60° in relation to the axis of the panel)

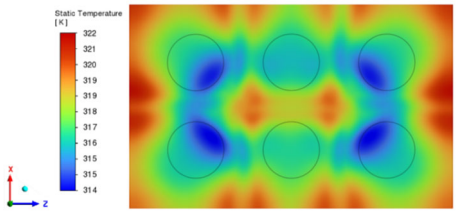
3.3. CFD Model Validation

The key step in the development of a numerical model is the validation of the results it provides. The model used for validation was a system assuming six fans blew air perpendicular to the panel surface. Additional experiments were conducted and compared to the simulation results (iterations 1–4). The relevant parameter in this analysis was the average temperature of the bottom surface of the panel (PVF film). The data collected are compared in Table 7 and on the graph shown in Figure 10. Validation was based on comparing the numerical model (CFD) value to the data obtained from a measurement series (EXP). It is worth noting that the temperature value obtained from the model results from a stationary simulation was not subject to any error; hence, it is a single value for each measurement series.

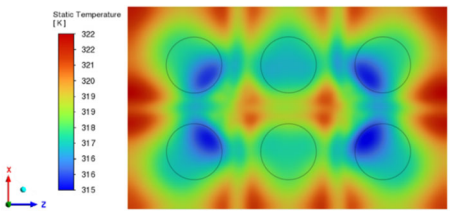
Table 7. Comparison of the results obtained from the validation experiment and the CFD model.

Iteration	The Average Temperature of Panel Surface (EXP) [K]	The Average Temperature of Panel Surface (CFD) [K]	Deviation of the CFD Model Results from the Experimental Value [K]
1	313.48 ± 2 K	317.64	−4.16
2	319.68 ± 2 K		2.04
3	314.45 ± 2 K		−3.19
4	315.12 ± 2 K		2.52
Average	315.68 ± 1.84 K *		1.96

* according to the evaluation of uncertainty.

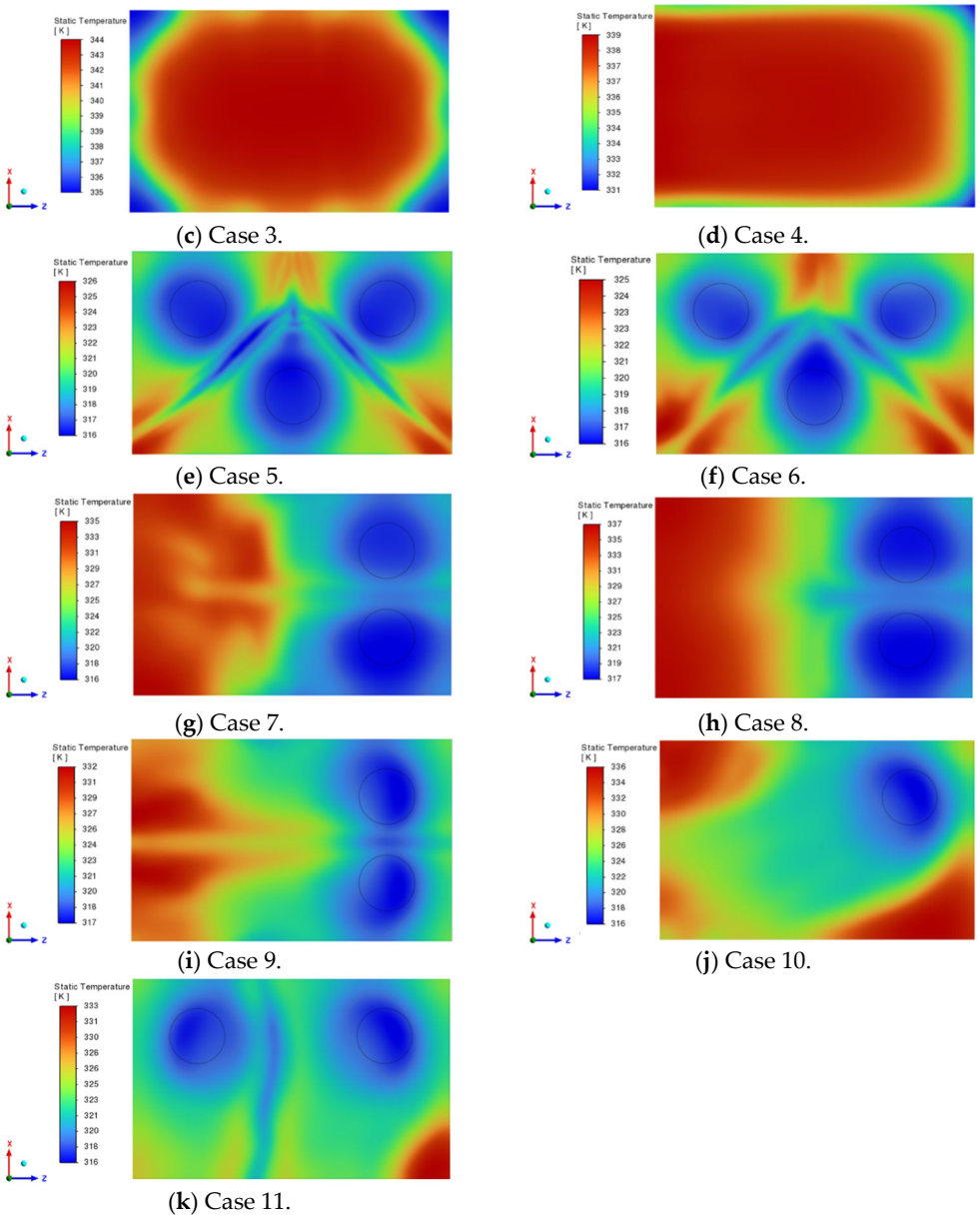


(**a**) Case 1.



(**b**) Case 2.

Figure 10. *Cont.*



(**c**) Case 3. (**d**) Case 4.

(**e**) Case 5. (**f**) Case 6.

(**g**) Case 7. (**h**) Case 8.

(**i**) Case 9. (**j**) Case 10.

(**k**) Case 11.

Figure 10. Temperature distribution of PV cells for Cases 1–11.

Based on the above analyses, the validity and accuracy of the numerical model can be confirmed. The result obtained from the numerical simulation slightly exceeds the range of the experimental values (~0.15 K). For the purposes of the following study, this deviation should be considered insignificant, and validation should be considered successful.

3.4. CFD Modeling Results

Based on the validated model, eleven cases were created, differing in the number of air inlets (fans) and their tilt in relation to the panel. It is worth noting that during the validation process, the average temperature of the bottom surface of the PV panel—PVF—was considered. In the case of the results presented below, these are the temperature values on the layer corresponding to the PV cells. In addition, the temperature distribution among the PV cells was analyzed in terms of its uniformity (see Figure 10). The panels inclined at an angle of 30° are intended to represent the actual conditions of standard PV installations, e.g., on the sloped roofs of residential buildings. However, the analyses carried out for the panels arranged horizontally have similarly high relevance. One reason against arranging PV panels in this way in practice is specifically the difficulty of their self-cooling. A solution

using fans can prove to be very helpful in these cases. The average temperature values collected in Table 8 can be directly related to the efficiency of the PV panel. This relationship is based on the temperature coefficient of the power $P_{\%} = -0.38\%/^{\circ}\mathrm{C}$ of the experimentally tested panel. This value can be entered into Equation (9) to determine the effect of different model configurations on the power of the panel under study.

$$\Delta P = P_{\%} \cdot P_R \cdot (T_{ave} - T_R) \quad (9)$$

where

ΔP—the variation in power related to the PV cell temperature [W],
P_R and T_R—the power [W] and average surface temperature [K] of the reference PV cells,
T_{ave}—the average temperature of the panels from the CFD model [K].

Table 8. Results obtained from the CFD model for each analized case.

Case Number	Average Temperature of PV Panel [K]	Power Difference [W]	Net Power Difference [W]
1 (Figure 10a)	318.50	2.89	−11.51
2 (Figure 10a)	318.67	2.87	−11.53
3 (Figure 10b)	341.72	−0.63 *	-
4 (Figure 10b)	337.58	0.00	0.00
5 (Figure 10c)	320.02	2.66	−4.54
6 (Figure 10c)	319.88	2.68	−4.52
7 (Figure 10d)	324.62	1.96	−2.84
8 (Figure 10d)	326.44	1.69	−3.11
9 (Figure 10d)	323.56	2.12	−2.68
10 (Figure 10e)	325.40	1.85	−0.55
11 (Figure 10f)	321.86	2.38	−2.42

* impact solely of panel tilt.

In the described analysis, the reference panel that was used to assess the effectiveness of the solutions from each variant was Case 4 because it is the one that assumed the conditions most similar to the panels installed in actual practice. It is worth mentioning that the PV panel power for this case was first determined based on the indications obtained from an experiment with parameters analogous to Case 3. Nevertheless, Table 8 presents the values in a way that allows for direct comparison to Case 4. It is important to recall that the single fan used in the experimental study had a power demand of P_{dem} = 2.4 W. This value was included in the calculation of net power gain (Equation (10)).

$$P_{nett} = \Delta P - P_{dem} \quad (10)$$

The obtained results allow for a number of correlations to be observed. Firstly, the influence of the tilt of the PV panel should be pointed out for the cases of cooling with three and six fans (Cases 1, 2, 5, and 6). Very close values are observed for the average temperature, and thus the power for this case. This implies a very minimal contribution of the natural convection process to heat transfer in these models. The other observations involve the cases where the fans were located exclusively in the lower part of the panel (Cases 7 and 8). In these options, some difference between the average temperature readings is visible (~2 K), indicating the influence of the angle of inclination on the operation of the panels for such variants. When considering all the cases, the highest power gain was observed for Cases 1 and 2 and 5 and 6. These were highly ineffective variants due to the high energy consumption of the equipment used. For this reason, it was decided to analyze using only one fan in Case 10. Moreover, the net power gain was also characterized by a negative value. Nevertheless, it was decided that further research should be carried out, which should eventually allow for appropriate conclusions to be established.

An important parameter is the angle of the airflow from the fans in relation to the panel's surface area. In Cases 9, 10, and 11, this was an angle of 60°. This treatment allowed

for relatively significant power gains to be achieved, while reducing the number of fans. Considering the case with a single fan, the power gain was unfortunately insufficient to cover the unit's demand. Nevertheless, it was not far behind the variants using a larger number of fans. This also indicates that further research on a single unit is required.

First, it is important to compare the cases containing the highest number of fans. Air damming between the fan locations is noticeable for Cases 1, 2, 3, and 4. This area is characterized by enhanced heat transfer, which is a positive aspect, but it may also indicate the sub-optimal, excessive output of the fans. Therefore, further research should be carried out to investigate the dependence of the fan parameters on their cooling capacity. The second relationship is related to adjusting the angle of air inflow over the panel surface. Cases 9, 10, and 11 show that the appropriate adjustment of the fan placement can provide a uniform temperature distribution over the panel. Figure 10a,b,e,f again shows the low impact of the panel tilt for the cases with a high number of fans. This influence is significant for their location on one side of the panel (Figure 10g,h,i,j). Case 8 represents the cooling of a tilted panel, whereas Case 7 represents a non-tilted panel. In this case, the inclined panel was characterized by a slight increase in temperature and a smaller cooling area. Therefore, the impact of these treatments should be carefully investigated in further analyses.

3.5. Parametric Analysis of the Most Prominent Case

The cases described above allowed for the selection of the most promising layout—Case 10 (Figure 10j). The use of a single fan directly translates into the lowest power demand. The fan position angle is one of the crucial parameters affecting the efficiency of the system. For this reason, further optimization analyses were carried out for Case 10.

This research used parametric analysis for varying fan tilt values in the vertical and horizontal axes. These were purely simulation studies using the experiment planning capabilities of the ANSYS Workbench software. It is worth noting that the other model parameters were fixed and analogous to those used in the previous simulations.

The plan of the experiment included two input parameters: the horizontal γ_{hor} and vertical γ_{vert} angles of the fan from the surface of the panel. For values of 0 degrees for both the parameters, the fan was oriented perpendicular to the panel, at $\gamma_{vert} = 90^{\circ}$ and $\gamma_{hor} = 0^{\circ}$ degrees, the airflow was parallel to the longer side of the panel, and for the contrary situation, it was parallel to the shorter one. The variation in these values was set to vary in the range 0–80°. It is worth mentioning that for $\gamma_{vert} = 0^{\circ}$, changing γ_{hor} had no effect on the simulation. As the output parameters, the average temperature of the photovoltaic cells T_{ave} and the range of temperature ΔT [K] of these cells were set. The second value was calculated with Equation (11).

$$\Delta T = T_{max} - T_{min} \tag{11}$$

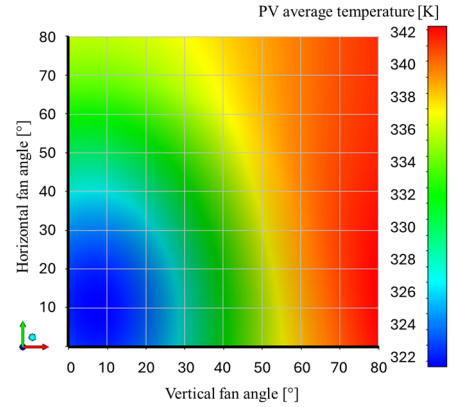
where T_{max} and T_{min} [K] are the highest and lowest temperature values on the surface of the PV cells, respectively.

A total of 44 simulations were carried out to select the appropriate response surfaces using the results. For this purpose, the Non-Parametric Regression method was used [30]. Dependency plots were obtained and are shown in Figures 11 and 12.

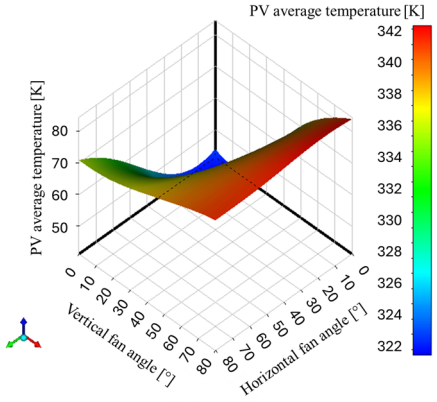
The created response surfaces allow for finding the optimal parameter values for which the lowest T_{ave} and ΔT readings can be observed. Analyzing the first plots (see Figure 10), it is possible to see a very strong influence of the fan orientation on the average panel temperature. The clear extremum of the function for $\gamma_{vert} \approx 7^{\circ}$ and $\gamma_{hor} \approx 10^{\circ}$ can be seen. The response surface indication of the effect of the slope on ΔT is unclear. It can be seen that the function has its minimum outside the study area under consideration. This is consistent with reality. When the fan is positioned parallel to the PV panel, the forced flow will remove heat from its surface at a minimal rate. The response surface also shows this regarding the average surface temperature. For high values of the fan angle, it takes the highest readings. Figure 12 shows that the function under study has a second local extreme for $\gamma_{vert} \approx 7^{\circ}$. The location of this minimum is similar to that observed in the first graph

(see Figure 11). It is worth noting that the effect of the horizontal fan angle on ΔT is much smaller than that of the vertical angle.

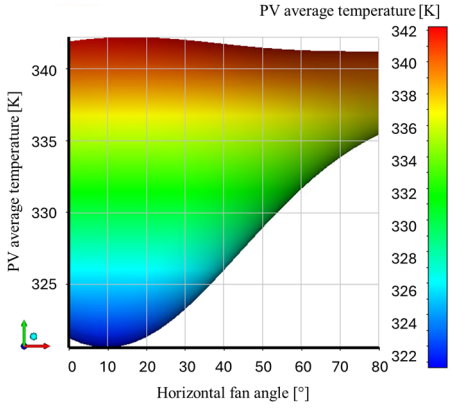
Reading the exact values from the substitute model resulted in the most optimal set of fan setting parameters, which are summarized in Table 9.



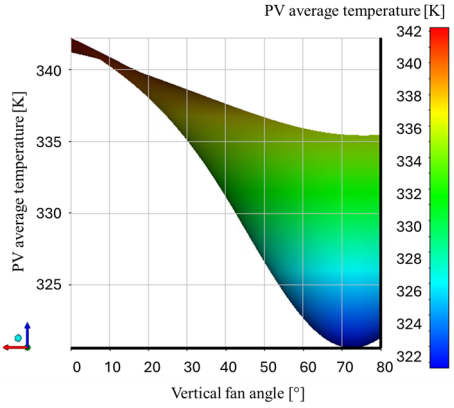
(**a**) Top view.



(**b**) Isometric view.

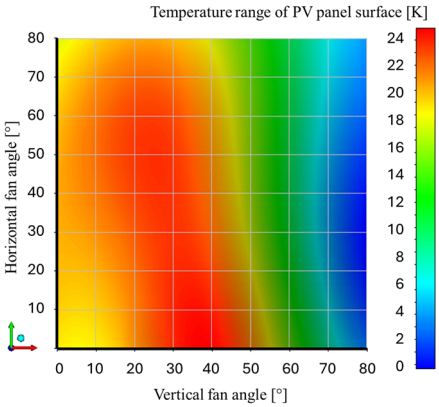


(**c**) Perpendicular to γ_{vert} axis.

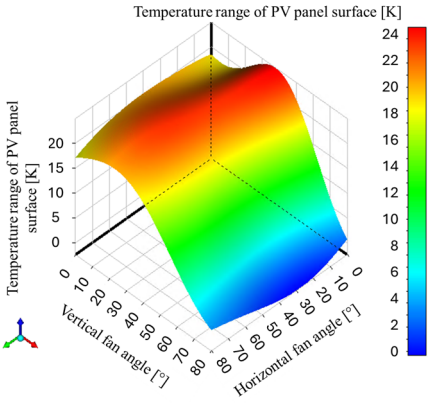


(**d**) Perpendicular to γ_{hor} axis.

Figure 11. Response surfaces showing the dependence of the fan setting on the average temperature of the PV panel surface.

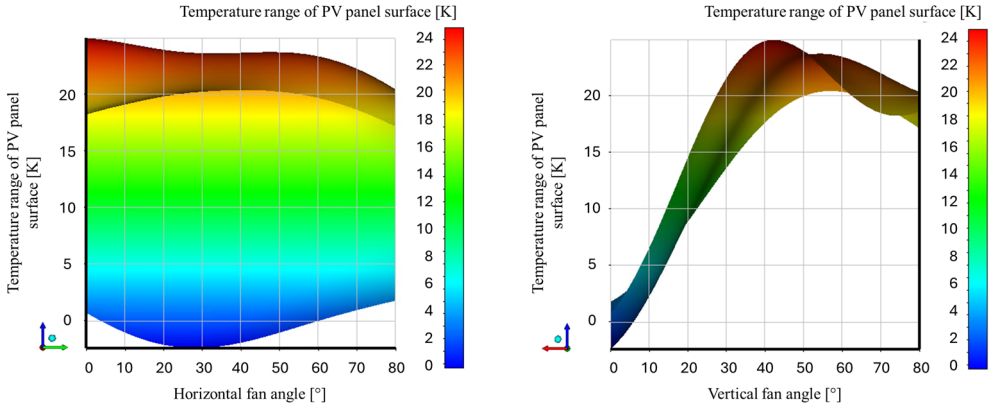


(**a**) Top view.



(**b**) Isometric view.

Figure 12. *Cont.*



(**c**) Perpendicular to γ_{vert} axis. (**d**) Perpendicular to γ_{hor} axis.

Figure 12. Response surfaces showing the dependence of the fan setting on the temperature range of the PV panel surface.

Table 9. The summary results of parametric analysis according to the substitute model.

Output Parameter	Value [K]	Inlet Parameter	Value [°]
Average temperature of PV panel surface T_{ave}	320.62	Fan angle in vertical axis γ_{vert}	7
		Fan angle in horizontal axis γ_{hor}	10
Temperature range at the surface of the PV panel ΔT	18.23	Fan angle in vertical axis γ_{vert}	7
		Fan angle in horizontal axis γ_{hor}	0

The substitute model obtained by the method described above may have some uncertainty alongside the numerical error of a single simulation. This is due to the use of an approximation method for creating the response surface. For this reason, an additional individual simulation was carried out with the parameters obtained from the substitute model. It was found that the more significant parameter in the analysis was the average surface temperature of the PV panel, so angles of $\gamma_{vert} = 7°$ and $\gamma_{hor} = 10°$ were applied for the airflow. The temperature distribution obtained from this analysis is presented in Figure 13.

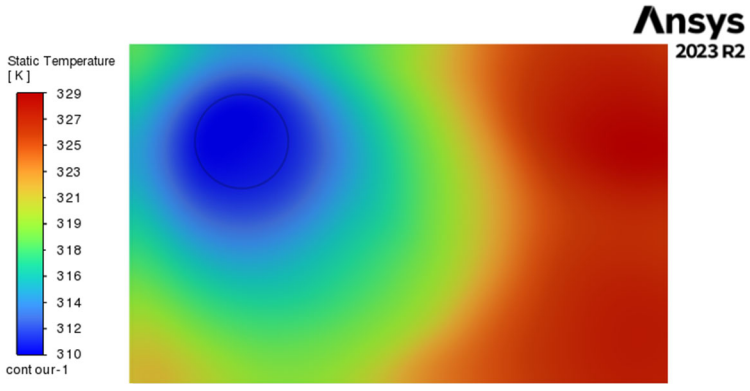


Figure 13. The temperature distribution for the model with optimal parameters.

Comparing the result of the optimum angle simulation and Case 10 (Figure 10j), it can be seen that the area of the lowest temperature values is more extensive for the larger fan angle. T_{min} and T_{max} have different values with a similar ΔT value. The simulation of the optimum model resulted in temperature values of $T_{ave} = 319.45$ K and $\Delta T = 20.66$ K. By relating the average temperature value to the reference case (Figure 11d), a value of $\Delta P = 3.77$ W was determined according to Equation (9). Considering the fan power

requirement, a net power gain of $\Delta P_{net} \approx 1.37$ W was obtained. This result directly indicates that the proposed solution is cost-effective.

During parametric analysis, the effect of different flow speeds on T_{ave} and ΔT was additionally investigated. It is worth pointing out that market research has shown that fans with similar dimensions and electrical power requirements can accelerate the airflow up to 3.0 m/s. For this reason, this analysis carried out assumed a variation in this parameter in the range of 0.5–3.0 m/s. The simulation results are shown in Figure 14.

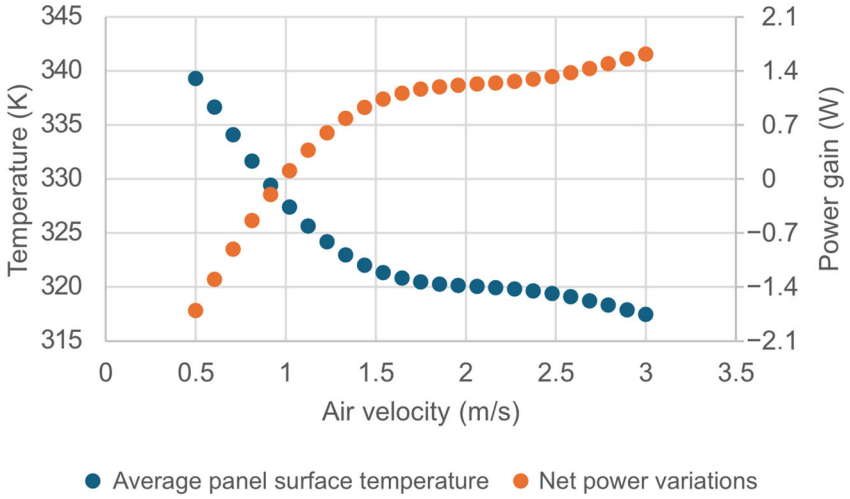


Figure 14. Dependence of air velocity on average temperature and net power gain of PV panel.

The presented graph shows that the dependence of flow velocity on T_{ave} and P_{net} has a declining nature throughout the range. It can be observed that for values higher than ~1.5 m/s, the impact of velocity changes is much smaller. In a situation where the cost of fans grows significantly as their effectiveness increased, seeking to implement devices to maximize this parameter may not be the optimal action from an economic point of view. Nevertheless, it is worth noting that the system starts to be effective after the velocity value exceeds only ~0.9 m/s. This may mean that using a fan with worse airflow acceleration capabilities and power requirements is possible and may show greater cost-effectiveness.

4. Conclusions

This paper presents a prototype of an air cooling system dedicated to PV panels. The proposed system is equipped with air fans mounted on a plate connected to the rear part of a 70 Wp photovoltaic panel. Different configurations of the system were tested, including different numbers of fans and different locations of the fans. Furthermore, a mathematical model was developed based on the results obtained, and simulations were carried out using the ANSYS Workbench 2023 R2 software. After the validation procedure, several configurations of the air cooling system were analyzed. The most important findings resulting from the experimental works are as follows:

- The maximum power generated in the PV panel under laboratory conditions varied from 47.08 W to 47.68 W when the average temperature of the PV panel surface was around 22.4–24.7 °C. This power dropped to 43.19 W and 40.09 W when the average temperature of the PV panel surface increased to approx. 50 °C and 60 °C, respectively.
- The maximum power values generated in the photovoltaic panel equipped with the developed cooling system with six air fans were approx. 45.06 W and 44.37 W in the same conditions (i.e., it was higher by 4.3–10.7% compared to that of the uncooled one, respectively).
- The difference between the considered variants is small when comparing the series with an installed cooling system. Consequently, it is possible to provide the efficient cooling of the tested PV panel using a low number of air fans.

Furthermore, the crucial findings resulting from numerical analysis are as follows:

- A low impact of the panel tilt was observed for the symmetrical distribution of fans as opposed to just laying them under a part of the PV panel.
- As the power requirement per fan was 2.4 W, the experimentally analyzed cases were not cost-effective, particularly the variant with six fans (net power difference = −11.51 W).
- The power gain values for all the variants were similar and ranged from 1.69 to 2.89 W; however, when considering the cooling system's power requirements, the variants using the smallest number of fans were clearly more promising.
- It was considered that the variant with a single fan inclined toward the surface of the PV panel, when properly optimized, could provide effective cooling in the tested system.
- The variant analysis of the numerical simulation results confirmed the above conclusions. It indicated that the vertical and horizontal tilts of the fan are one of the most important parameters for maximizing the system's efficiency.
- Parametric analysis allowed for the selection of the optimal fan tilt settings ($\gamma_{vert} \approx 7^{\circ}$ and $\gamma_{hor} \approx 10^{\circ}$), which translated into the cost-effective operation of the proposed system (net power increase of ~3.1%).
- Additional analysis of the effect of the velocity of air exiting the fan on power gain was carried out, and the minimum value (0.9 m/s) for which the system is viable was determined.

Additionally, it should also be noted that the above experimental studies were carried out on a laboratory scale, using low-power PV panels and an artificial light source. These studies allowed for the creation of a mathematical model of the proposed system and the determination of the possibility of increasing the efficiency of PV panels using active air cooling systems. Parametric analysis allowed for the tilt angle to be optimized in the vertical and horizontal planes. Further research will be carried out on full-size PV panels under real conditions. This will verify the mathematical model and allow for the validity of this type of system for full-scale PV installations. In addition, it will allow for the economic analysis of the proposed system.

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References

1. Deng, X.; Lv, T.; Meng, X.; Li, C.; Hou, X.; Xu, J.; Wang, Y.; Liu, F. Assessing the carbon emission reduction effect of flexibility option for integrating variable renewable energy. *Energy Econ.* **2024**, *132*, 107461. [CrossRef]
2. Salau, A.O.; Alitasb, G.K. MPPT efficiency enhancement of a grid connected solar PV system using Finite Control set model predictive controller. *Heliyon* **2024**, *10*, e27663. [CrossRef]
3. Available online: https://www.rynekelektryczny.pl/moc-zainstalowana-fotowoltaiki-w-polsce/ (accessed on 1 July 2024).
4. Seapan, M.; Hishikawa, Y.; Yoshita, M.; Okajima, K. Temperature and irradiance dependences of the current and voltage at maximum power of crystalline silicon PV devices. *Sol. Energy* **2020**, *204*, 459–465. [CrossRef]
5. Al-Amri, F.; Saeed, F.; Mujeebu, M.A. Novel dual-function racking structure for passive cooling of solar PV panels –thermal performance analysis. *Renew. Energy* **2022**, *198*, 100–113. [CrossRef]
6. Hasanuzzaman, M.; Malek, A.B.M.A.; Islam, M.M.; Pandey, A.K.; Rahim, N.A. Global advancement of cooling technologies for PV systems: A review. *Sol. Energy* **2016**, *137*, 25–45. [CrossRef]

7. Sato, D.; Yamada, N. Review of photovoltaic module cooling methods and performance evaluation of the radiative cooling method. *Renew. Sustain. Energy Rev.* **2019**, *104*, 151–166. [CrossRef]
8. Mariam, E.; Ramasubramanian, B.; Reddy, V.S.; Dalapati, G.K.; Ghosh, S.; Pa, T.S.; Chakrabortty, S.; Motapothula, M.R.; Kumar, A.; Ramakrishna, S.; et al. Emerging trends in cooling technologies for photovoltaic systems. *Renew. Sustain. Energy Rev.* **2024**, *192*, 114203. [CrossRef]
9. Odeh, S.; Behnia, M. Improving Photovoltaic Module Efficiency Using Water Cooling. *Heat Transf. Eng.* **2009**, *30*, 499–505. [CrossRef]
10. Sornek, K.; Goryl, W.; Figaj, R.; Dąbrowska, G.; Brezdeń, J. Development and Tests of the Water Cooling System Dedicated to Photovoltaic Panels. *Energies* **2022**, *15*, 5884. [CrossRef]
11. Sornek, K.; Goryl, W.; Głowacka, A. Improving the performance of photovoltaic panels using a direct water cooling system. *J. Sustain. Dev. Energy, Water Environ. Syst.* **2023**, *11*, 1110468. [CrossRef]
12. Alshibil, A.M.A.; Farkas, I.; Víg, P. Sustainability contribution of hybrid solar collector towards net-zero energy buildings concerning solar cells wasted heat. *Energy Sustain. Dev.* **2023**, *74*, 185–195. [CrossRef]
13. Nahar, A.; Hasanuzzaman, M.; Rahim, N.A. A Three-Dimensional Comprehensive Numerical Investigation of Different Operating Parameters on the Performance of a Photovoltaic Thermal System With Pancake Collector. *J. Sol. Energy Eng.* **2017**, *139*, 031009. [CrossRef]
14. Lekbir, A.; Hassani, S.; Ab Ghani, M.R.; Gan, C.K.; Mekhilef, S.; Saidur, R. Improved energy conversion performance of a novel design of concentrated photovoltaic system combined with thermoelectric generator with advance cooling system. *Energy Convers. Manag.* **2018**, *177*, 19–29. [CrossRef]
15. Arnaoutakis, G.E.; Katsaprakakis, D.A. Energy Yield of Spectral Splitting Concentrated Solar Power Photovoltaic Systems. *Energies* **2024**, *17*, 556. [CrossRef]
16. Zhang, H.; Yue, H.; Huang, J.; Liang, K.; Chen, H. Experimental studies on a low concentrating photovoltaic/thermal (LCPV/T) collector with a thermoelectric generator (TEG) module. *Renew. Energy* **2021**, *171*, 1026–1040. [CrossRef]
17. Duan, J. The PCM-porous system used to cool the inclined PV panel. *Renew. Energy* **2021**, *180*, 1315–1332. [CrossRef]
18. Micheli, L.; Fernandez, E.F.; Almonacid, F.; Reddy, K.S.; Mallick, T.K. Enhancing ultra-high CPV passive cooling using least-material finned heat sinks. *AIP Conf. Proc.* **2015**, *1679*, 27. [CrossRef]
19. Tina, G.M.; Scavo, F.B.; Micheli, L.; Rosa-Clot, M. Economic comparison of floating photovoltaic systems with tracking systems and active cooling in a Mediterranean water basin. *Energy Sustain. Dev.* **2023**, *76*, 101283. [CrossRef]
20. Brinkworth, B.J.; Cross, B.M.; Marshall, R.H.; Yang, H. Thermal regulation of photovoltaic cladding. *Sol. Energy* **1997**, *61*, 169–178. [CrossRef]
21. Elbreki, A.M.; Muftah, A.F.; Sopian, K.; Jarimi, H.; Fazlizan, A.; Ibrahim, A. Experimental and economic analysis of passive cooling PV module using fins and planar reflector. *Case Stud. Therm. Eng.* **2021**, *23*, 100801. [CrossRef]
22. Patil, M.; Sidramappa, A.; Shetty, S.K.; Hebbale, A.M. Experimental study of solar PV/T panel to increase the energy conversion efficiency by air cooling. *Mater. Today Proc.* **2023**, *92*, 309–313. [CrossRef]
23. Elminshawy, N.A.S.; El Ghandour, M.; Gad, H.M.; El-Damhogi, D.G.; El-Nahhas, K.; Addas, M.F. The performance of a buried heat exchanger system for PV panel cooling under elevated air temperatures. *Geothermics* **2019**, *82*, 7–15. [CrossRef]
24. Hussien, A.; Eltayesh, A.; El-Batsh, H.M. Experimental and numerical investigation for PV cooling by forced convection. *Alex. Eng. J.* **2023**, *64*, 427–440. [CrossRef]
25. Wang, W.; Laumert, B. Simulate a 'Sun' for Solar Research: A Literature Review of Solar Simulator Technology. *Eng. Environ. Sci.* **2014**. Available online: https://api.semanticscholar.org/CorpusID:106667773 (accessed on 6 August 2024).
26. Sher, A.A.; Ahmad, N.; Sattar, M.; Ghafoor, U.; Shah, U.H. Effect of Various Dusts and Humidity on the Performance of Renewable Energy Modules. *Energies* **2023**, *16*, 4857. [CrossRef]
27. Bickler, D. The simulation of solar radiation. *Sol. Energy* **1962**, *6*, 64–68. [CrossRef]
28. ANSYS Workbench 23R2 Help Files, n.d. Available online: https://link.springer.com/chapter/10.1007/0-387-30623-4_5#citeas (accessed on 15 June 2024).
29. Zhou, J.; Yi, Q.; Wang, Y.; Ye, Z. Temperature distribution of photovoltaic module based on finite element simulation. *Sol. Energy* **2015**, *111*, 97–103. [CrossRef]
30. Nonparametric Regression. In *All of Nonparametric Statistics*; Springer Texts in Statistics; Springer: New York, NY, USA, 2006; pp. 61–123. [CrossRef]

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